

NORILSK AP, Russian Federation

WMO: 319811

Lat: 69.311N

Lon: 87.332E

Elev: 574

StdP: 14.39

Time Zone: 7.00 (E07)

Period: 04-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-40.4	-36.5	-48.9	0.3	-40.9	-43.8	0.4	-36.3	44.5	-2.5	39.5	-1.0	9.8	50	1.027

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	14.1	77.2	60.5	73.2	59.4	68.2	57.3	63.0	72.2	60.9	69.7	58.4	67.2	12.0	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
59.4	77.3	66.1	57.0	70.8	63.9	54.0	63.3	61.4	28.9	70.6	27.2	70.2	25.5	67.3	71.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 37.1	33.5	29.9	-46.1	81.7	6.4	4.7	-50.7	85.1	-54.4	87.9	-58.0	90.5	-62.6	93.9		
(5)			-46.2	66.2	6.3	3.3	-50.8	68.6	-54.4	70.5	-58.0	72.4	-62.6	74.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	18.6	-8.6	-12.1	-2.6	11.7	24.5	47.1	57.6	49.7	40.2	19.9	0.3	-6.9	(6)
(7)		DBStd	26.83	12.66	14.30	15.74	14.50	9.99	9.28	8.41	7.23	7.82	11.33	12.55	14.40	(7)
(8)		HDD50	11905	1817	1738	1631	1150	795	165	21	94	306	933	1491	1764	(8)
(9)		HDD65	16986	2282	2158	2096	1600	1256	543	263	477	743	1398	1941	2229	(9)
(10)		CDD50	437	0	0	0	0	4	77	257	85	14	0	0	0	(10)
(11)		CDD65	41	0	0	0	0	0	5	34	2	0	0	0	0	(11)
(12)		CDH74	286	0	0	0	0	0	36	240	10	0	0	0	0	(12)
(13)		CDH80	38	0	0	0	0	0	3	35	0	0	0	0	0	(13)
(14)	Wind	WSAvg	14.3	17.5	14.9	15.1	15.0	14.0	12.4	12.4	12.0	12.9	14.1	14.2	16.7	(14)
(15)	Precipitation	PrecAvg	34.3	2.8	2.5	2.4	2.4	2.4	2.3	2.4	3.1	2.7	4.3	3.3	3.4	(15)
(16)		PrecMax	43.7	5.7	4.7	5.2	4.8	4.1	6.3	5.3	5.6	7.0	6.4	5.6		(16)
(17)		PrecMin	17.0	0.5	0.5	0.4	0.1	0.5	0.3	0.4	1.1	1.2	1.3	0.7	0.3	(17)
(18)		PrecStd	5.5	1.3	1.1	1.2	1.1	0.9	1.0	1.3	0.9	1.0	1.5	1.4	1.3	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	23.2	22.7	30.4	41.0	66.3	78.8	84.0	75.3	66.6	49.6	32.3	23.3	(19)
(20)			MCWB	23.1	22.4	29.9	35.6	53.1	60.2	62.9	61.2	56.8	46.9	32.3	23.3	(20)
(21)		2%	DB	15.7	17.8	25.1	37.8	44.9	73.1	80.4	69.8	59.0	39.5	28.5	20.8	(21)
(22)			MCWB	15.4	17.5	24.6	34.8	40.4	57.6	61.6	58.9	51.5	38.7	28.3	20.5	(22)
(23)		5%	DB	12.1	12.3	21.1	35.6	39.5	66.6	76.7	64.6	55.1	37.1	22.8	17.2	(23)
(24)			MCWB	11.7	12.1	20.4	33.4	36.2	53.6	60.6	55.8	49.2	36.4	22.0	16.5	(24)
(25)		10%	DB	7.1	6.8	17.5	32.2	37.2	62.3	71.8	60.8	51.5	33.8	17.8	12.3	(25)
(26)			MCWB	6.6	6.3	16.4	31.1	34.6	51.8	59.9	54.9	47.8	33.2	16.8	11.9	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	22.9	22.0	29.7	36.7	52.6	62.7	67.7	62.7	58.2	47.2	32.2	23.2	(27)
(28)			MCD B	23.1	22.2	30.3	39.0	66.1	74.5	77.2	71.8	65.8	48.8	32.2	23.2	(28)
(29)		2%	WB	14.7	17.4	24.8	35.1	40.3	58.8	64.4	60.1	53.7	39.0	28.3	19.9	(29)
(30)			MCD B	14.9	17.8	25.2	37.4	44.0	69.8	73.2	67.3	57.1	39.3	28.3	20.1	(30)
(31)		5%	WB	11.3	12.3	20.1	33.3	36.7	55.5	62.6	57.5	50.2	35.9	21.8	16.3	(31)
(32)			MCD B	11.6	12.3	20.7	35.0	38.9	65.1	71.6	62.9	53.0	36.3	22.6	16.8	(32)
(33)		10%	WB	6.3	6.2	16.4	31.0	34.5	52.1	60.7	55.4	47.9	32.8	16.8	12.0	(33)
(34)			MCD B	6.7	6.6	17.3	32.6	36.4	60.6	69.8	59.9	51.1	33.2	17.8	12.4	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	10.6	11.5	12.9	12.8	10.3	13.3	14.1	11.7	9.3	8.2	10.6	10.9	(35)
(36)			MCDBR	13.5	15.1	12.2	9.9	11.5	21.0	18.5	16.8	13.5	6.8	13.1	14.2	(36)
(37)			MCWBR	13.4	15.2	12.2	8.3	8.7	10.8	8.6	8.9	8.5	6.1	13.3	13.2	(37)
(38)		5% WB	MCDBR	13.5	15.3	12.5	9.2	10.8	19.1	16.1	14.7	11.8	6.6	13.6	13.5	(38)
(39)			MCWBR	13.4	15.4	12.6	8.2	8.5	12.0	8.9	8.9	8.5	6.3	13.8	13.5	(39)
(40)	Clear-Sky Solar Irradiance	taub		0.212	0.247	0.303	0.328	0.359	0.374	0.402	0.394	0.326	0.249	0.199	N/A	(40)
(41)		taud		1.847	1.953	1.988	2.023	2.054	2.305	2.262	2.233	2.356	2.063	1.874	N/A	(41)
(42)		Ebn at Noon		6	148	215	251	257	259	245	229	218	147	5	0	(42)
(43)		Edh at Noon		2	15	29	38	42	34	35	32	21	13	1	0	(43)
(44)	All-Sky Solar Radiation	RadAvg		4	113	608	1295	1804	1696	1590	994	505	160	18	0	(44)
(45)		RadStd		1	15	65	106	148	148	224	113	57	30	2	0	(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (4 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page